

Pancreas

Headings

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Introduction

- soft, lobulated and elongated organ
- partly exocrine and partly endocrine

Functions

1. Exocrine function:
 - secrete alkaline pancreatic juice that contains Bicarbonates & Pancreatic enzymes
2. Endocrine function:
 - Islets of Langerhans secrete insulin, glucagon, somatostatin

Location

- lies across posterior abdominal wall
- T12 to L3 level
- Retroperitoneal (tail is intraperitoneal)
- Occupies epigastric, umbilical and left hypochondriac regions

Dimensions

- J-shaped or retort shaped
 - thickness: 1.5 to 2 cm
 - width: 2 to 4 cm
 - length: 12 to 15 cm
 - weight: 80 to 90 gm

Parts

- Divided into 4 parts: Head, Neck, Body & Tail
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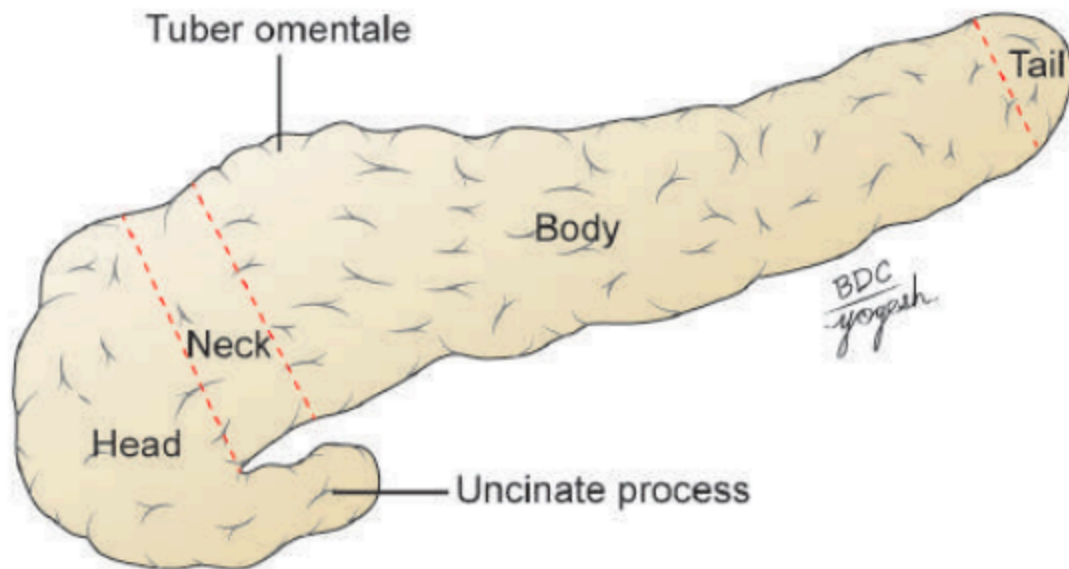


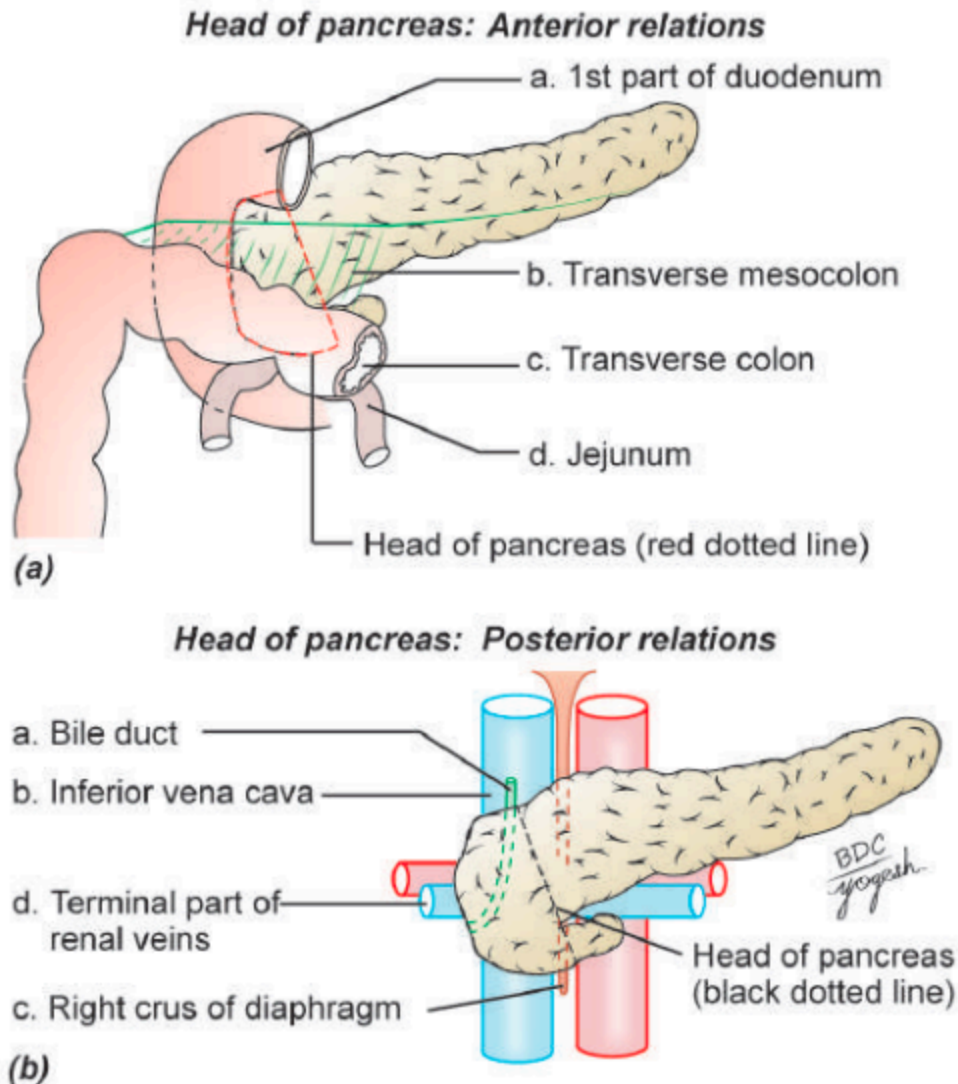
Fig. 23.12: Parts of pancreas

Relations

- Head:
 - enlarged flattened right end of pancreas, situated within the 'C-shaped' curve of the duodenum
 - The head of the pancreas has the following features:
 - a. Three borders: Superior, inferior and right-lateral
 - b. Two surfaces: Anterior and posterior
 - c. One process: Uncinate process.
 - Relations of the Head:

- Three borders:
 - a. Superior border is related to:
 - First part of the duodenum
 - Superior pancreaticoduodenal artery.
 - b. Inferior border is related to:
 - 3rd part of the duodenum
 - Inferior pancreaticoduodenal artery.
 - c. Right lateral border is related to:
 - 2nd part of the duodenum
 - Terminal part of the bile duct
 - Anastomosis between the two pancreaticoduodenal arteries.
- Two surfaces:
 - a. Anterior surface is related, from above downwards, to:
 - a. 1st part of duodenum
 - b. Transverse colon
 - c. Jejunum which is separated from it by peritoneum
 - b. Posterior surface is related to:
 - a. Inferior vena cava
 - b. Terminal parts of the renal veins
 - c. Right crus of the diaphragm
 - d. Bile duct which runs downwards and to the right and is often embedded in the substance of pancreas.
- Uncinate process:
 - The uncinate process is a triangular, hook-like projection from lower part of the head. It is related:
 - a. Anteriorly to the superior mesenteric vessels and
 - b. Posteriorly to the abdominal aorta.

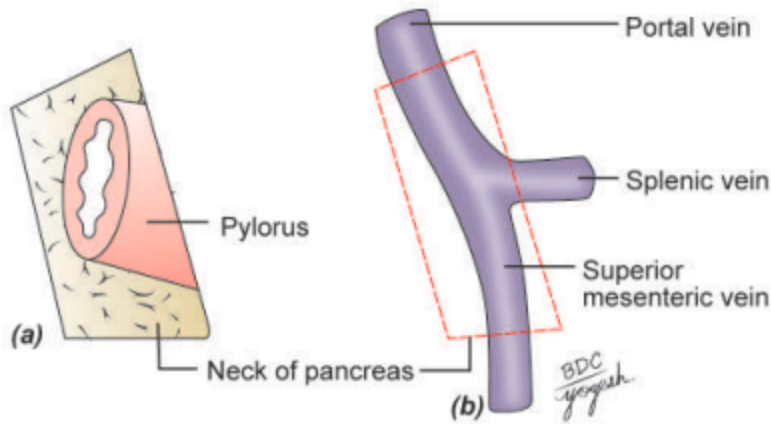
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Figs 23.14a and b: Relations of head of the pancreas: (a) Anterior; (b) Posterior

- Neck:
 - slightly constricted part of the pancreas between its head and body
 - The neck has the following features:
 - a. Two surfaces: Anterior and posterior
 - b. Two borders: Upper and lower.
 - Relations of Neck:
 - a. Anterior surface is related to: The peritoneum covering the posterior wall of the lesser sac, and the pylorus.
 - b. Posterior surface is related to the termination of the superior mesenteric vein and the beginning of the portal vein
 - c. Upper border is related to the 1st part of duodenum.
 - d. Lower border is related to the root of transverse mesocolon.

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Figs 23.15a and b: Relations of the neck of the pancreas:
(a) Anterior and (b) posterior

- Body:
 - elongated & extends from its neck to the tail
 - The body of pancreas is slightly triangular in crosssection. It has:
 - a. Two surfaces: Anterior and posterior
 - b. Two borders: Superior and inferior
 - c. One process: Tuber omentale: It is a rounded conical elevation from the right end of the superior border
 - Relations of Body:
 - Superior border
 - related to coeliac trunk over the tuber omentale and hepatic artery to the right and splenic artery to the left of tuber omentale.
 - Inferior border
 - related to the superior mesenteric vessels at its right end.
 - Anterior surface
 - covered by peritoneum and gives attachment to the transverse mesocolon
 - Relations above the attachment of transverse mesocolon:
 - a. Lesser sac
 - b. Stomach
 - Relations below the attachment of transverse mesocolon:
 - a. Infracolic compartment of greater sac
 - b. Coils of jejunum
 - c. Duodenojejunal flexure
 - d. Left colic flexure.
 - Posterior surface
 - devoid of the peritoneum. It lies on fusion fascia of Toldt. It is related to:
 - a. Aorta

- b. Origin superior mesenteric artery
- c. Left suprarenal gland
- d. Upper pole of left kidney
- e. Left renal vein
- f. Splenic vein (may be embedded in the gland)
- g. Left crus of diaphragm.

- Tails:

- narrowest and the most left lateral portion of the pancreas
- lies in the splenorenal (lienorenal) ligament along with splenic vessels
- It is the only mobile part of pancreas.
- It contains the most significant number of islets of Langerhans.

Duct System

- The exocrine pancreas is drained by two ducts—main and accessory
 - Main Pancreatic Duct of Wirsung
 - lies near the posterior surface of the pancreas and is recognised easily by its white colour
 - Within the head of the pancreas, the pancreatic duct is related to the bile duct which lies on its right side.
 - The two ducts enter the wall of the 2nd part of the duodenum and join to form the hepatopancreatic ampulla of Vater which opens by a narrow mouth on the summit of the major duodenal papilla, 8 to 10 cm distal to the pylorus.
 - Accessory Pancreatic Duct of Santorini
 - begins in the lower part of the head, crosses the front of the main duct with which it communicates and opens into the duodenum at the minor duodenal papilla.
 - The papilla of accessory pancreatic duct is situated 6 to 8 cm distal to the pylorus.

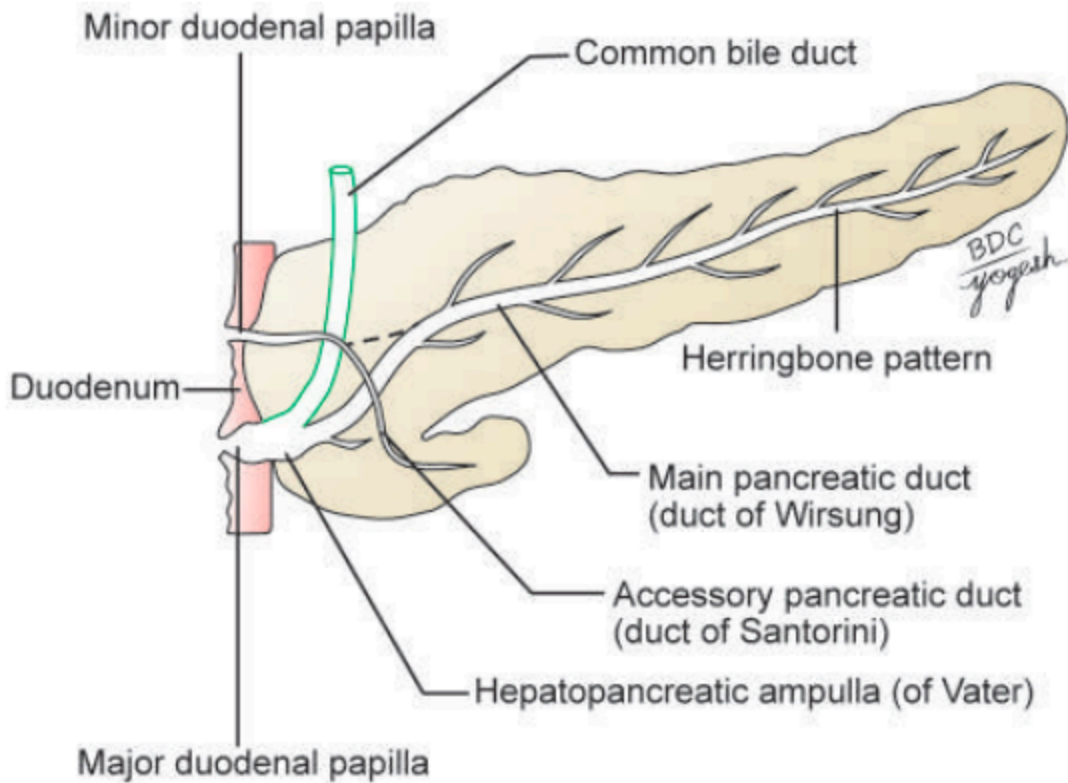


Fig. 23.17: The pancreatic ducts

Blood supply

- Arterial Supply:
 - i. Splenic artery – branch of coeliac trunk.
 - ii. Superior pancreaticoduodenal artery – branch of gastroduodenal artery.
 - iii. Inferior pancreaticoduodenal artery – branch of superior mesenteric artery.

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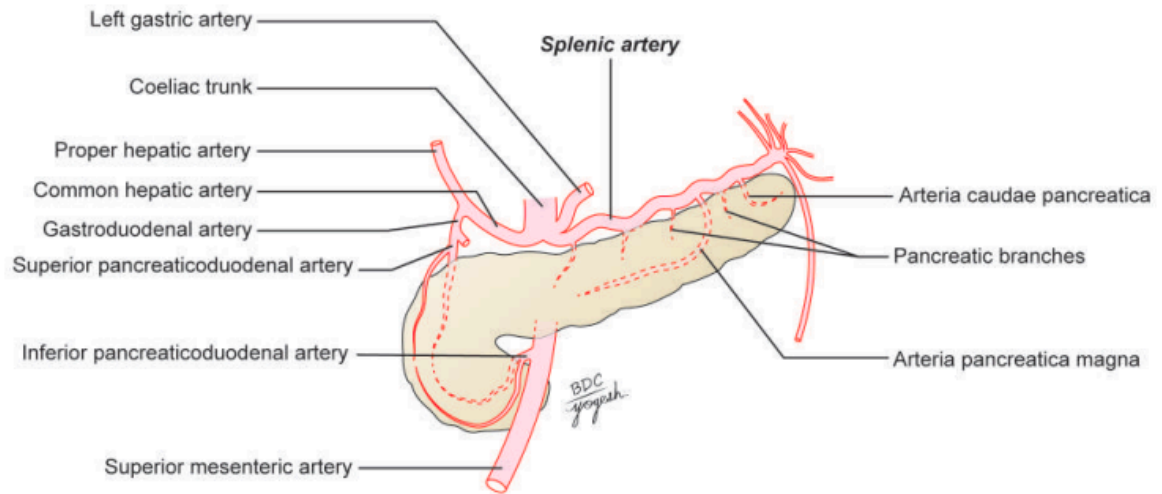


Fig. 23.18: Arterial supply of the pancreas

- Venous Supply:
 - i. Portal vein
 - ii. Superior mesenteric vein
 - iii. Splenic vein.

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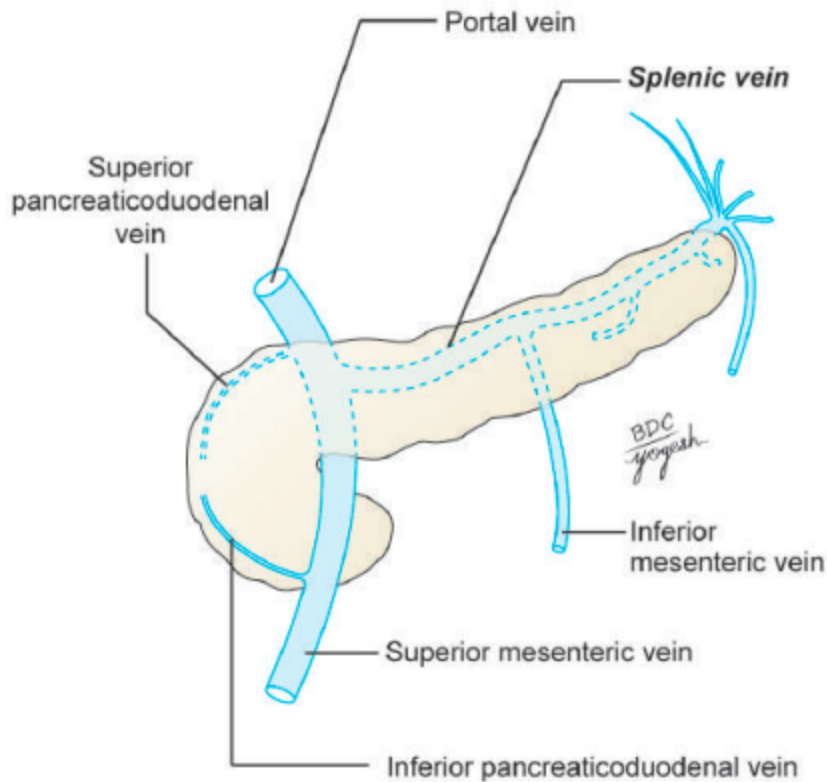


Fig. 23.19: Venous drainage of the pancreas

Lymphatic Drainage

- pancreas drain into the following lymph nodes:

- i. Pancreaticosplenic nodes [main lymphatic nodes]
- ii. Coeliac nodes
- iii. Superior mesenteric nodes
- iv. Pyloric nodes.

Nerve Supply

1. Parasympathetic fibres from vagus nerve:
 - These are secretomotor fibres
2. Sympathetic fibres from coeliac and superior mesenteric plexuses:
 - These are vasomotor.

Clinical

- Carcinoma of head:
 - may compress bile duct that leads to obstructive jaundice.
 - may also cause duodenal obstruction and compression of portal vein (resulting in ascites and portal hypertension).
- Acute pancreatitis:
 - an acute inflammation of pancreas that has severe complications and high mortality (death) rate.
 - It occurs because of abnormal activation of pancreatic proteolytic enzymes in the pancreas (usually, these enzymes get activated in the intestine).
 - Activated enzymes cause inflammation, oedema and cellular death.
- Chronic pancreatitis:
 - long-standing inflammation of pancreas (repeated attacks of mild acute pancreatitis).
- Pain from pancreatitis:
 - is poorly localized.
 - Pain is referred to epigastrium.
 - Pain is also referred to posterior paravertebral region and around the lower thoracic vertebrae, due to inflammation of soft tissues of retroperitoneum.
 - Their afferents are being sent through lower intercostal nerves.
- Pseudocyst of pancreas:
 - is a circumscribed collection of fluid containing pancreatic enzymes, blood and necrotic tissue.
 - It is located in lesser sac and may present symptoms similar to pancreatitis.
 - It occurs as a complication of pancreatitis.